PROTOCOL PRIMITIVE

GULLIANT

Verifiable User State Primitive
for Programmable Interoperability

PROBLEM

User state in Web3 is not portable and cannot be securely migrated.

- Permissions and tiers are tied to wallet addresses
- Protocol state cannot be transferred between accounts
- Every protocol rebuilds the same state logic

RESULT

Composability is broken because identity is tied to the signing key, not protocol state.

SOLUTION

A primitive for verifiable migration of protocol-controlled user state.

- Append-only user activity logs
- Deterministic hash-chain state
- Protocol-authorized export
- Replay-safe migration

`Replay protection via single-use authorization`

State becomes portable without losing trust

WHY IT MATTERS

Web3 lacks a standard user state layer. Gulliant enables:

PERSISTENCE

Maintain user state across wallet changes

COMPOSABILITY

Reusable state across protocols

SECURITY

Verifiable migration without trust assumptions

This is a primitive, not an application.

TECHNICAL IMPLEMENTATION

- On-chain enforced invariants
- SHA-256 hash-chain model
- Append-only architecture
- Snapshot-based migration
- Replay protection via single-use authorization
- Comprehensive integration tests
- Solana Devnet deployment
- Verifiable build (SHA-256 match)

DEPLOYMENT

PROGRAM ID

```
DyQAWvMytpgRiE2T9bhzSFCe9b9actobfHFB7RVReX7g
```

NETWORK

Solana Devnet

VERIFICATION

Deployed binary matches local build (SHA-256)